



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

考试科目: Signals and Systems

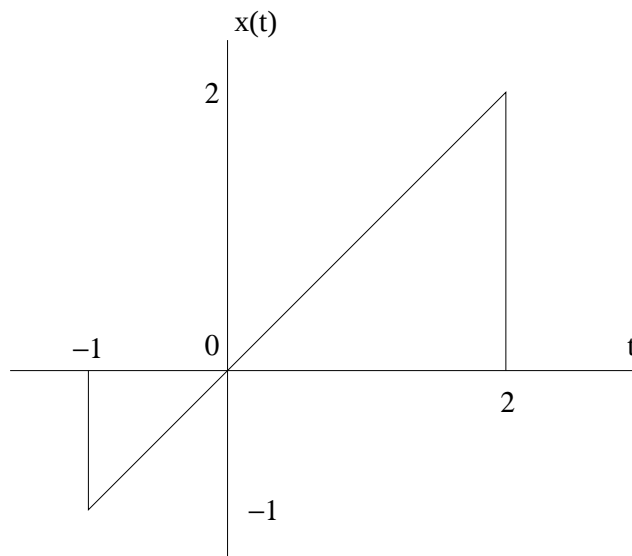
课程号: EE205

开课单位: 电子系

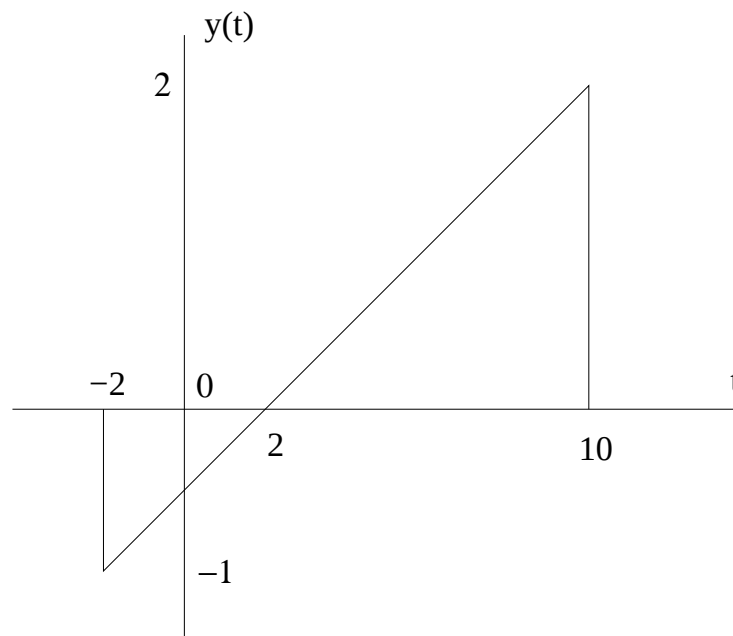
命题教师: 刘德荣

题号	1	2	3	4	5	6	7	8	9	10
分值	10	10	10	10	10	10	10	10	10	10

1. A signal $x(t)$ is given as in the figure.



Carefully sketch $y(t) = x\left(\frac{t-2}{4}\right)$.



2 points for $t = -2$.
 3 points for $t = 0$.
 3 points for $t = 2$.
 2 points for $t = 10$.

2. Compute $y[n] = x[n] * h[n]$ using the definition of convolution, where

$$x[n] = \begin{cases} 1, & 1 \leq n \leq 3, \\ 0, & \text{otherwise,} \end{cases}$$

$$h[n] = \begin{cases} 1, & n = 1, \\ 2, & n = 2, \\ 2, & n = 3, \\ 1, & n = 4, \\ 0, & \text{otherwise,} \end{cases}$$

By definition, we have

$$\begin{aligned} y[n] &= x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{k=1}^3 x[k]h[n-k] \\ &= x[1]h[n-1] + x[2]h[n-2] + x[3]h[n-3]. \end{aligned}$$

That is to say

$$y[n] = h[n-1] + h[n-2] + h[n-3].$$

Therefore,

n	$y[n]$	points
1	0	1
2	1	1
3	3	1
4	5	2
5	5	2
6	3	1
7	1	1
8	0	1

3. A continuous-time signal $x(t)$ is known to be real and periodic with period $T = 4$. In its Fourier series representation, there are only five nonzero coefficients and they are given by

$$a_0 = 2, \quad a_1 = j, \quad a_{-1} = -j, \quad a_3 = a_{-3} = 1.$$

Determine an expression for $x(t)$.

From Fourier series representation, we have

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = a_0 + a_1 e^{j\omega_0 t} + a_{-1} e^{-j\omega_0 t} + a_3 e^{j3\omega_0 t} + a_{-3} e^{-j3\omega_0 t},$$

where $\omega_0 = 2\pi/T = \pi/2$.

We therefore have

$$\begin{aligned} x(t) &= 2 + j e^{j(\pi/2)t} - j e^{-j(\pi/2)t} + e^{j3(\pi/2)t} + e^{-j3(\pi/2)t} \\ &= 2 - 2 \frac{e^{j\pi t/2} - e^{-j\pi t/2}}{2j} + 2 \cos \frac{3\pi}{2} t \\ &= \underbrace{2}_{3 \text{ points}} - \underbrace{2 \sin \frac{\pi}{2} t}_{4 \text{ points}} + \underbrace{2 \cos \frac{3\pi}{2} t}_{4 \text{ points}}. \end{aligned}$$

Answer correct without j : 10 points.

Answer correct with j : Give 6 points total.

4. Use the definition of Fourier transform to calculate the Fourier transform of

$$x(t) = e^{-2t}u(t-3).$$

Determine the magnitude and the phase of this Fourier transform.

$$\begin{aligned} X(j\omega) &= \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt = \int_3^{\infty} e^{-2t}e^{-j\omega t}dt = \frac{1}{-2-j\omega}e^{-2t-j\omega t}\Big|_3^{\infty} \\ &= \frac{e^{-(2+j\omega)3}}{2+j\omega}. \quad (4 \text{ points}) \end{aligned}$$

The magnitude is given by

$$|X(j\omega)| = \frac{e^{-6}}{\sqrt{4+\omega^2}} \quad (3 \text{ points})$$

and the phase is

$$\angle X(j\omega) = -3\omega - \tan^{-1} \frac{\omega}{2}. \quad (3 \text{ points})$$

5. Determine the time signal corresponding to the following Fourier transform

$$X(e^{j\omega}) = \frac{5e^{-j\omega}}{1 - \frac{1}{6}e^{-j\omega} - \frac{1}{6}e^{-2j\omega}}.$$

We have

$$X(e^{j\omega}) = \frac{5e^{-j\omega}}{(1 - \frac{1}{2}e^{-j\omega})(1 + \frac{1}{3}e^{-j\omega})} = \frac{3e^{-j\omega}}{1 - \frac{1}{2}e^{-j\omega}} + \frac{2e^{-j\omega}}{1 + \frac{1}{3}e^{-j\omega}}.$$

$$x[n] = \underbrace{3 \left(\frac{1}{2}\right)^{n-1} u[n-1]}_{5 \text{ points}} + \underbrace{2 \left(-\frac{1}{3}\right)^{n-1} u[n-1]}_{5 \text{ points}}.$$

6. A causal LTI system is given by the following differential equation

$$\frac{d^2y(t)}{dt^2} + 22\frac{dy(t)}{dt} + 40y(t) = 120x(t).$$

- Determine the frequency response of the system.
- Sketch the Bode (magnitude and phase) plots of the system.

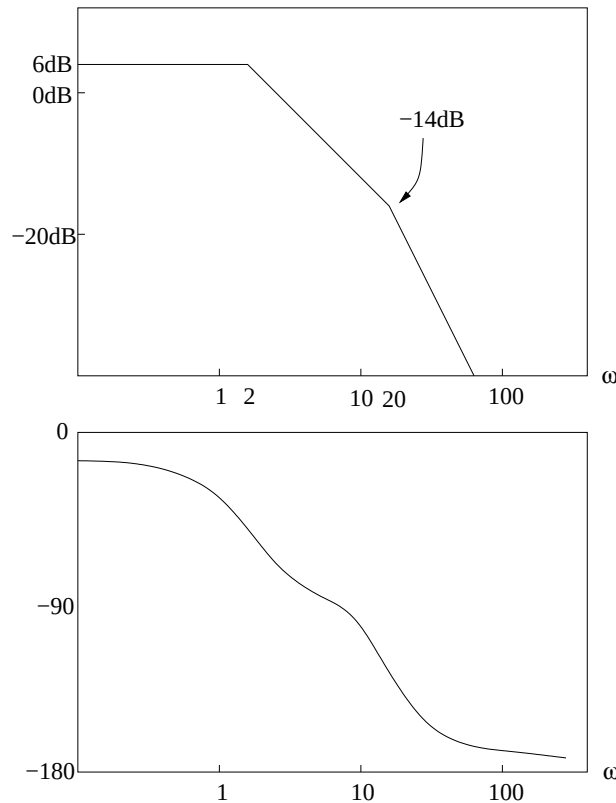
(a) The frequency response is

$$H(j\omega) = \frac{120}{(j\omega)^2 + j22\omega + 40}. \quad (3 \text{ points})$$

(b) We have

$$H(j\omega) = \frac{120}{(j\omega + 2)(j\omega + 20)} = 3 \left(\frac{1}{1 + \frac{j\omega}{2}} \right) \left(\frac{1}{1 + \frac{j\omega}{20}} \right).$$

The Bode plots are shown.



(ω small, 6dB) → 1 point.

Approximating (2, 6dB) → 1 point.

Approximating (20, -14dB) → 1 point.

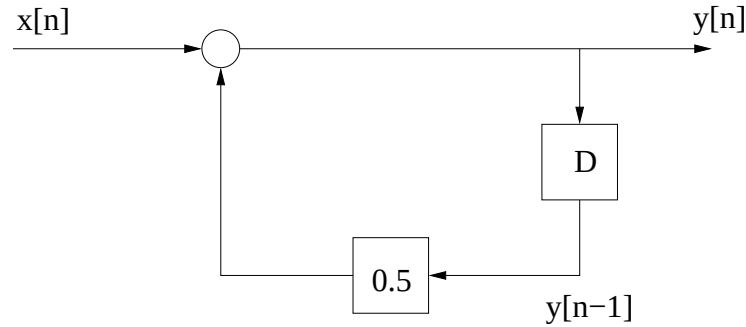
Going down at -40dB/decade for large ω → 1 point.

(ω small, 0 degrees) → 1 point.

(ω = 2 ~ 20, -90 degrees) → 1 point.

(ω large, -180 degrees) → 1 point.

7. A first-order linear discrete-time system is given in the figure.



Determine the output $y[n]$ when the input is $x[n] = \left(\frac{1}{3}\right)^n u[n]$.

From the figure, we obtain the difference equation for this system as

$$y[n] - \frac{1}{2}y[n-1] = x[n].$$

The frequency response can be obtained as

$$H(e^{j\omega}) = \frac{1}{1 - \frac{1}{2}e^{-j\omega}}. \quad (2 \text{ points})$$

The Fourier transform of the input signal is

$$X(e^{j\omega}) = \frac{1}{1 - \frac{1}{3}e^{-j\omega}}. \quad (2 \text{ points})$$

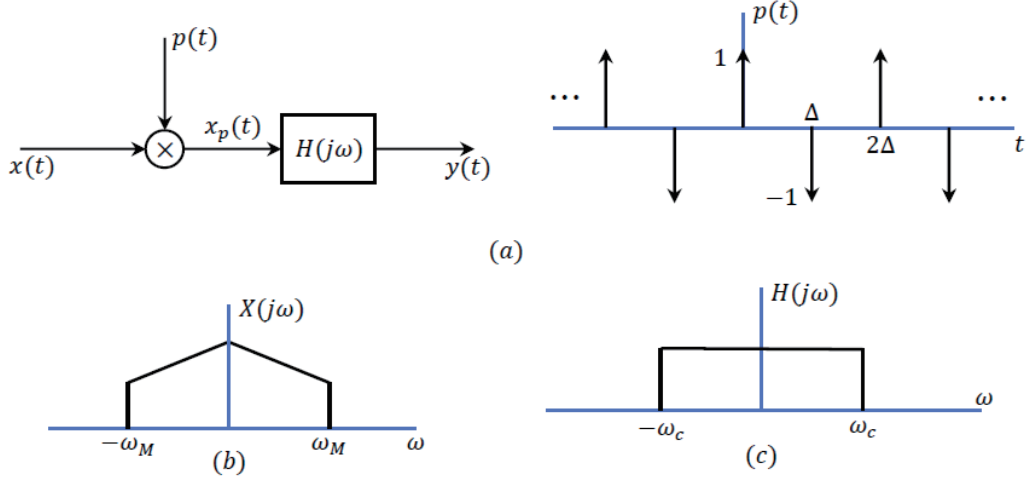
Therefore

$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega}) = \frac{1}{(1 - \frac{1}{2}e^{-j\omega})(1 - \frac{1}{3}e^{-j\omega})} = \frac{3}{1 - \frac{1}{2}e^{-j\omega}} - \frac{2}{1 - \frac{1}{3}e^{-j\omega}},$$

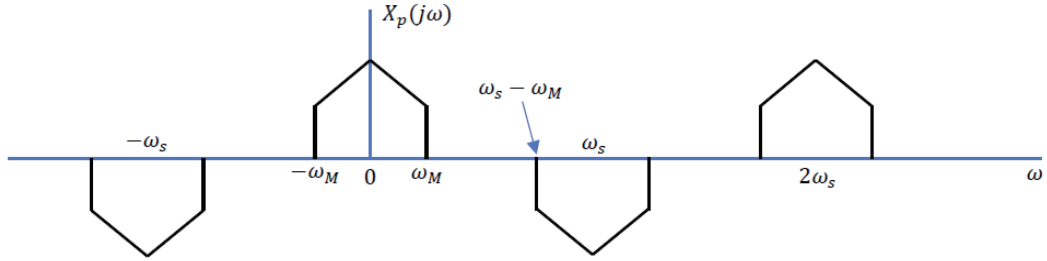
and

$$y[n] = \underbrace{3 \left(\frac{1}{2}\right)^n u[n]}_{3 \text{ points}} - \underbrace{2 \left(\frac{1}{3}\right)^n u[n]}_{3 \text{ points}}.$$

8. Shown in the Figure (a) is a system in which the sampling signal is an impulse train with alternating sign. The Fourier transform of the input signal is given in Figure (b). $H(j\omega)$ is a lowpass filter used to recover $x(t)$ from $x_p(t)$. Determine the condition for Δ in terms of ω_M for which $x(t)$ can be recovered perfectly from $x_p(t)$, i.e., $y(t) = x(t)$. What is the range of ω_c for the perfect recovery of $x(t)$?



The Fourier transform of $x_p(t)$ is sketched as follows:



From this we can see that the conditions for the perfect recovery of $x(t)$ from $x_p(t)$ are:

$$\omega_s = \frac{2\pi}{\Delta},$$

$$\omega_s - \omega_M > \omega_M,$$

$$\omega_M \leq \omega_c \leq \omega_s - \omega_M.$$

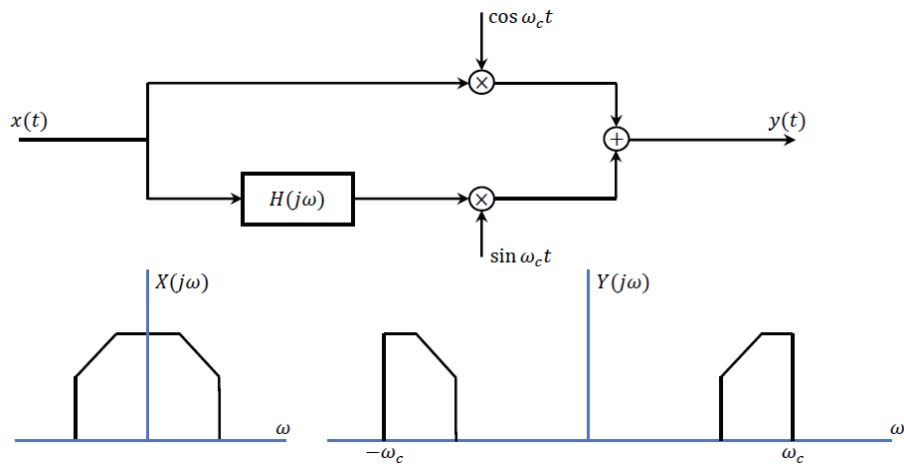
Therefore, it is required

$$\Delta < \frac{\pi}{\omega_M} \quad (6 \text{ points})$$

and the choice of ω_c must satisfy

$$\omega_M \leq \omega_c \leq \omega_s - \omega_M. \quad (4 \text{ points})$$

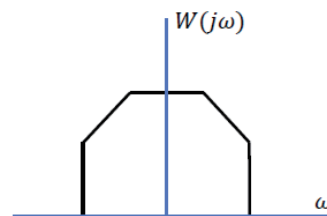
9. Shown in the figure is a single-sideband amplitude modulation system in which $y(t)$ requires half the transmission bandwidth comparing to the double-sideband modulation. Design a synchronous demodulator to recover $x(t)$ from $y(t)$.



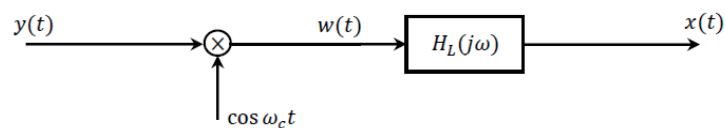
Multiply $y(t)$ with $\cos \omega_c t$, we obtain

$$w(t) = y(t) \cos \omega_c t.$$

The Fourier transform of $w(t)$ is sketched as follows:



Therefore, a synchronous demodulator is given by:



where $H_L(j\omega)$ is an ideal lowpass filter.

7 points for multiplying with $\cos \omega_c t$.

3 points for lowpass filter.

10. For $y(t) = h(t) * x(t)$, we are given $y(t) = e^{-t}u(t)$ and $x(t) = e^{-2t}u(t)$. Determine $h(t)$.

We have

$$Y(j\omega) = \frac{1}{1+j\omega}, \quad (1 \text{ point})$$

$$X(j\omega) = \frac{1}{2+j\omega}. \quad (1 \text{ point})$$

Therefore,

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{2+j\omega}{1+j\omega} = 1 + \frac{1}{1+j\omega} \quad (2 \text{ points}) \Rightarrow h(t) = \underbrace{\delta(t)}_{3 \text{ points}} + \underbrace{e^{-t}u(t)}_{3 \text{ points}}.$$